

Deep LLM Usage Analysis

A Data-Driven Profile of 2.7 Years of Large Language Model Engagement

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Executive Summary

This document presents a comprehensive, data-driven analysis of 2,398 conversations and 71,427 messages exchanged across ChatGPT and Claude over 33 months (August 2023 – April 2026). All figures derive from complete official platform exports processed through reproducible open-source scripts. This analysis complements the separately published *Quantifying LLM Practice Hours* methodology paper; it does not duplicate that work.

Key findings:

- **21 distinct model versions used** across the full GPT-3.5 through GPT-5.4 (Thinking) lineage, plus reasoning models (o1, o3) and Deep Research — demonstrating continuous adoption of each new model generation within weeks of release.
- **17 tool and feature types** actively used, including web search (473 conversations), code interpreter (383), extended reasoning (283), vision/multimodal (246), and image generation (72). 331 conversations used custom GPTs across 35 unique templates.
- **Prompt engineering technique adoption grew measurably** over the 33-month period, with constraint-setting rising from near-zero to 11.7% of messages, context front-loading from zero to 7.3%, and code inclusion from near-zero to 5.9% on Claude.
- **A decisive platform migration** from ChatGPT to Claude occurred between December 2025 and March 2026, with Claude’s monthly share rising from 15% to 93% in four months.
- **272 conversations (11.3%) link directly to named projects** including a major AI project (81 conversations), VCAT (Voynich Manuscript data, 28), The Muser (music AI orchestration, 3), and academic coursework (45).

Metric	ChatGPT	Claude	Combined
Conversations	1,835	563	2,398
Messages	50,412	21,015	71,427

Metric	ChatGPT	Claude	Combined
User words written	1,330,677	730,740	2,061,417
Assistant words generated	9,414,865	3,567,147	12,982,012
Tokens exchanged	17,026,808	6,365,947	23,392,755
Active days	847	292	883
Date range	2023-08 to 2026-04	2024-03 to 2026-04	2023-08 to 2026-04

For context: the combined word output exceeds 12.9 million words — equivalent to approximately 130 full-length novels. The 883 unique active days represent usage on 88% of all days across the 33-month period.

1. Usage at Scale

1.1 Volume and Trajectory

Usage grew through three distinct phases:

Phase 1 — Early Exploration (Aug 2023 – Jan 2024). ChatGPT-only. Moderate volume averaging 380 messages/month across 37 conversations/month. Short conversations (median 8 messages). Characteristic of learning the tool’s capabilities.

Phase 2 — Ramp and Dual-Platform (Feb 2024 – Nov 2025). ChatGPT usage intensified, peaking at 4,203 messages in April 2025. Claude adopted in March 2024 and used concurrently at 5–25% share. Combined monthly volume averaged 3,400 messages. Conversations grew longer (median rising to 16+ messages on Claude).

Phase 3 — Claude-Primary Intensive (Dec 2025 – Apr 2026). Claude surged from 44% share in December 2025 to 93% in March 2026. February 2026 was the single most active month: 7,339 combined messages, with Claude accounting for 6,117 (83%). This period corresponds to intensive work on the major system architecture projects.

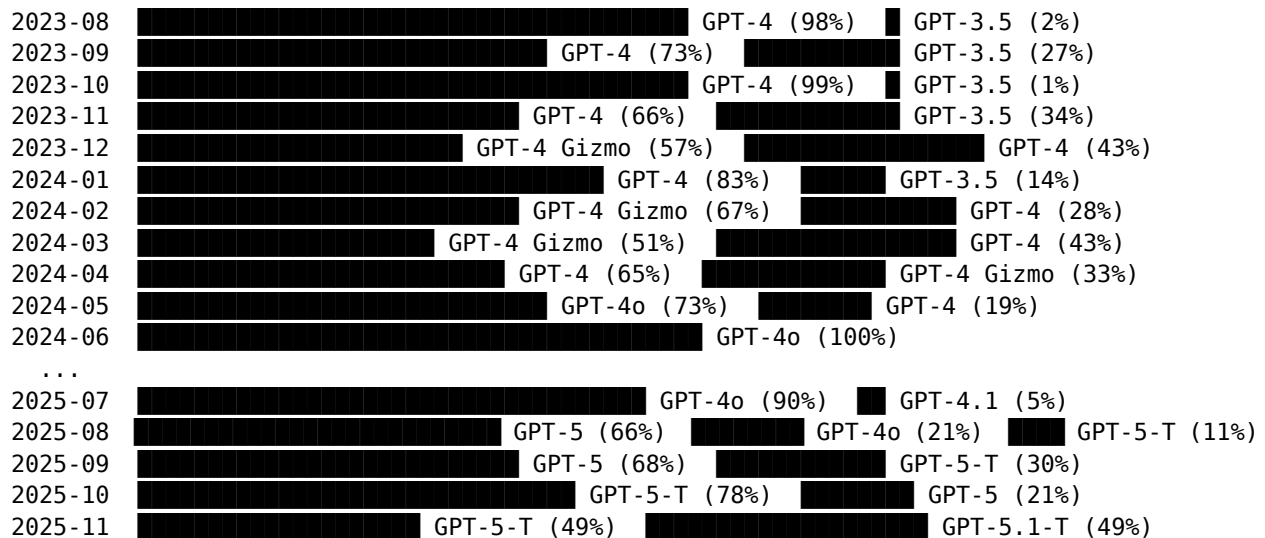
Activity patterns show peak usage hours between 21:00–06:00 UTC (evening through late night), with the lowest activity at 12:00–16:00 UTC. Tuesday and Wednesday are the most active days (12,974 and 11,863 messages respectively), approximately 50% busier than weekends.

1.2 Model Usage and Technology Adoption

The ChatGPT export contains per-message model metadata, revealing usage across 21 distinct model versions:

Model	First Used	Last Used	Messages	Conversations
GPT-4	2023-08-08	2024-05-25	2,057	309
GPT-3.5 Turbo	2023-08-11	2024-05-25	308	61
GPT-4 (Custom GPTs)	2023-12-01	2024-04-09	1,093	78
GPT-4o	2024-05-15	2025-09-11	17,218	983
o1	2025-02-20	2025-02-20	3	1
GPT-4.5	2025-03-05	2025-07-21	95	8
GPT-4.1	2025-06-19	2025-08-07	171	11
o3	2025-07-16	2025-07-16	48	1
GPT-5	2025-07-22	2026-01-07	3,089	117
GPT-5 (Thinking)	2025-08-09	2025-11-12	4,070	87
GPT-5 Turbo Mini	2025-09-13	2025-10-16	37	7
GPT-5.2	2025-10-06	2026-03-04	888	47
GPT-5.2 (Thinking)	2025-10-06	2026-03-23	8,188	134
GPT-5.3	2025-10-06	2026-04-17	338	19
GPT-5 Instant	2025-11-09	2025-11-12	13	3
GPT-5.1	2025-11-12	2025-11-19	18	5
GPT-5.1 (Thinking)	2025-11-12	2025-12-11	1,980	50
GPT-5.1 Instant	2025-11-12	2025-12-06	105	7
GPT-5.2 Instant	2025-12-21	2026-03-14	39	5
Deep Research	2026-01-18	2026-01-18	14	2
GPT-5.4 (Thinking)	2026-03-14	2026-04-16	381	11

Model Migration Timeline



2025-12	
2026-01	
2026-02	
2026-03	
2026-04	

The migration pattern reveals consistent early adoption: each major model version was adopted within days to weeks of public availability. GPT-4o dominated the mid-period (17,218 messages, 983 conversations), while GPT-5.2 (Thinking) became the primary model for the later intensive period (8,188 messages, 134 conversations).

Reasoning model investment. Combined usage of thinking/reasoning model variants (GPT-5 Thinking, GPT-5.1 Thinking, GPT-5.2 Thinking, GPT-5.4 Thinking, o1, o3) totals 14,670 messages — 37.5% of all ChatGPT assistant messages. This indicates sustained investment in extended reasoning capabilities, not casual experimentation.

Feature and Tool Adoption

Feature	Conversations	% of ChatGPT Total
Web Search	473	25.8%
Code Interpreter	383	20.9%
Extended Reasoning	283	15.4%
Multimodal/Vision	246	13.4%
Memory	137	7.5%
Image Generation (DALL-E)	72	3.9%
File Browser	39	2.1%
File Search	32	1.7%
Canvas	13	0.7%
Deep Research	2	0.1%

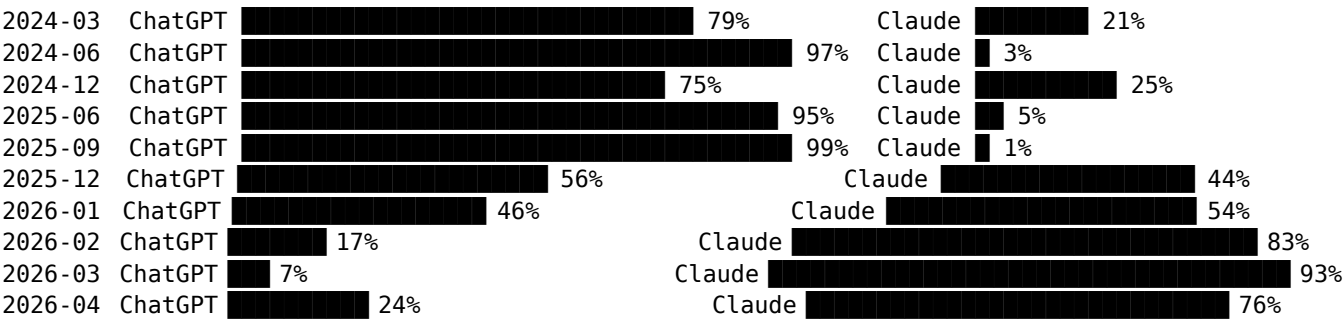
Custom GPTs. 331 conversations (18.0% of ChatGPT total) used custom GPTs across 35 unique configurations. This indicates active use of the custom GPT ecosystem for specialized workflows, not just the default interface.

Regeneration and Branching

213 conversations (11.6%) contain at least one branch point — a node in the conversation tree with multiple children, indicating the user regenerated a response or explored an alternative path. Total branch points: 341. This provides a direct behavioral signal of selective dissatisfaction with model outputs.

1.3 Platform Allocation

Claude was adopted in March 2024, initially as a secondary platform at 5–25% of monthly volume. The shift to Claude-primary happened rapidly over four months:



The platforms serve different usage profiles. Claude conversations average 39.8 messages (median 16) versus ChatGPT’s 27.5 (median 8). The 90th percentile conversation depth is 104 messages on Claude versus 64 on ChatGPT. Claude is used for deeper, more sustained work.

Claude also has 22 named projects with descriptions, prompt templates, and attached documents — a structured project management approach not available in ChatGPT’s interface. This organizational infrastructure likely contributed to the platform shift.

2. Domain Portfolio

2.1 Topic Distribution

Conversations were classified across 17 domain categories using keyword matching against titles, with content-based fallback for ambiguous titles. Classification confidence varies by category (see Methodology).

Category	Conversations	% of Total
AI/ML/LLM	200	8.3%
Research/Knowledge	195	8.1%
Software Engineering	165	6.9%
Personal/Life	153	6.4%
Domain-Specific Processing	149	6.2%
Writing/Creative	120	5.0%
Systems/Hardware/DevOps	115	4.8%

Category	Conversations	% of Total
Academic/Coursework	113	4.7%
Visual/Design	96	4.0%
Career/Professional	88	3.7%
Music/Audio	87	3.6%
Finance/Business	40	1.7%
Gaming/Entertainment	34	1.4%
Mythology/Philosophy	31	1.3%
Military/Strategy	30	1.2%
Voynich/Cryptography	26	1.1%
Legal	13	0.5%

1,113 conversations (46.4%) did not match any category keywords and are classified as “Other.” This reflects the limitations of keyword-based classification on short or colloquial titles (e.g., “Quick Question,” “Excitement for Pasta Wine”). The classified 53.6% covers the topically distinctive conversations with reasonable confidence (93.1% agreement in validation sampling).

The domain portfolio demonstrates breadth across technical (software engineering, systems, AI/ML), creative (writing, music, visual design), academic (coursework, research), and applied (domain-specific processing, career) domains. This is not single-domain usage.

2.2 Project Attribution

272 conversations (11.3%) were linked to specific named projects using keyword matching against known project names and descriptions. Confidence levels reflect match quality: “high” means the project name appears explicitly in the conversation title; “medium” means 2+ keyword matches; “low” means a single keyword or thematic match.

Project	Conversations	High Confidence	Medium	Low
Domain Pipeline Project	81	39	4	38
Academic coursework	45	0	5	40
Hardware / systems build	39	0	13	26
VCAT / Voynich	28	12	3	13
Freelance tech writing	15	1	1	13
Audio Project	13	0	0	13
Personal writing project	12	5	3	4
Personal Branding	9	1	1	7
Creative writing project	8	1	1	6

Project	Conversations	High Confidence	Medium	Low
Software project	4	0	3	1
Personal project A	4	0	1	3
Personal research project	4	0	1	3
The Muser	3	2	0	1
Aesthetics	3	0	0	3
Personal project B	2	0	1	1

Domain Pipeline Project is the dominant project by conversation count, with 39 high-confidence attributions. This aligns with the separately documented a major AI project status report showing 100,000+ lines of Python, hundreds of processed works, and millions of structured records — built primarily through LLM-directed development sessions.

Note on attribution completeness. Many project-related conversations have ambiguous titles (e.g., a Claude conversation titled “JSON parsing fix” might relate to a major AI project but lacks identifying keywords). The 272 linked conversations represent a lower bound; actual project-related usage is likely higher. Claude project-to-conversation linkage is done entirely through keyword matching because the platform export does not include direct project-conversation associations.

3. Prompt Engineering Effectiveness

3.1 Technique Adoption

Ten prompt engineering techniques were tracked across all 35,225 user messages. Results are reported as adoption rates (percentage of user messages using each technique), not as composite scores.

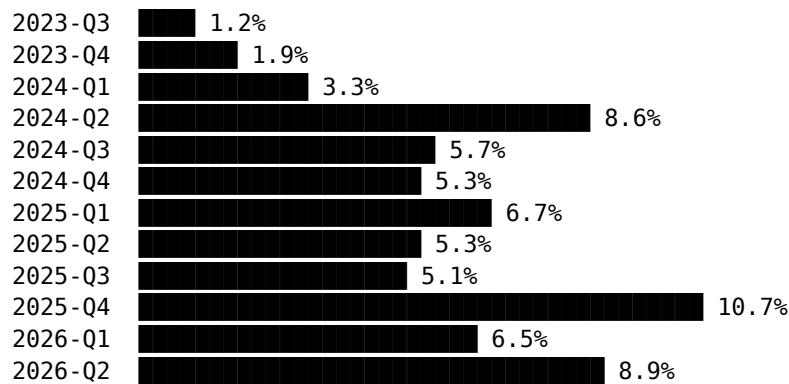
Technique	Overall	ChatGPT	Claude	First Appeared
Constraint specification	6.5%	6.0%	7.7%	2023-08
Context front-loading (>200 words)	4.6%	3.5%	7.2%	2024-02
Iterative refinement signals	3.3%	2.9%	4.3%	2024-03
Code inclusion	2.7%	1.4%	5.9%	2023-10
Multi-step instructions	1.7%	1.4%	2.3%	2023-11
Examples given	1.3%	1.0%	2.0%	2024-03
Role/persona assignment	0.4%	0.3%	0.4%	2024-06
Output format specification	0.3%	0.3%	0.3%	2025-03

Technique	Overall	ChatGPT	Claude	First Appeared
Meta-prompting	0.1%	0.1%	0.1%	—
Prior-conversation reference	0.1%	0.1%	0.1%	—

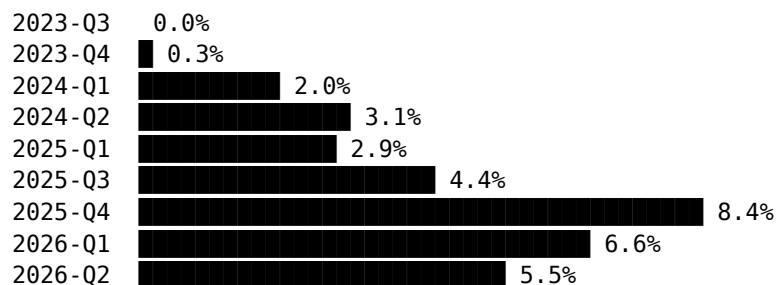
Platform differential. Every technique except role assignment and output formatting shows meaningfully higher adoption on Claude. Context front-loading (7.2% vs 3.5%) and code inclusion (5.9% vs 1.4%) are particularly skewed toward Claude, consistent with using Claude for deeper, more technical work.

Technique Growth Over Time (Quarterly)

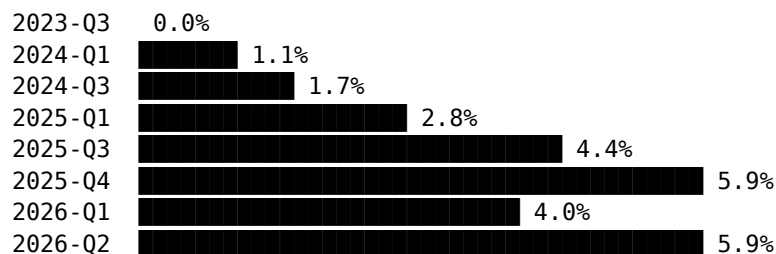
Constraint specification:



Context front-loading (>200 words per message):



Iterative refinement signals:



All three techniques show clear upward trajectories, with notable inflection points in Q4 2025 coinciding with the platform shift to Claude and the intensification of systems-level project work.

Interpretation. These percentages appear low because the denominator includes all user messages — including short follow-ups, acknowledgments, and conversational turns within deep sessions. In conversations where the user is actively engineering a prompt (first messages, complex requests), the rates would be substantially higher. The value of these numbers is in their *relative* comparison across platforms and over time, not their absolute levels.

3.2 Effectiveness Evidence

Conversation Outcomes

Of 1,575 conversations with 5+ messages:

Outcome	Count	Percentage
Standard completion	1,292	82.0%
Frustrated (3+ corrections or explicit frustration)	148	9.4%
Converged (explicit positive feedback)	107	6.8%
Neutral	28	1.8%

“Standard completion” indicates the conversation ended with an assistant response and no explicit positive or negative signal — the normal case where the user got what they needed and moved on. Only “converged” and “frustrated” represent strong outcome signals. The 6.8% explicit convergence rate and 9.4% frustration rate yield a **positive-to-negative outcome ratio of 0.72**, which must be interpreted carefully: most successful conversations simply end without explicit feedback.

Platform comparison. Claude shows both higher explicit convergence (12.1% vs 5.0%) and higher frustration (14.6% vs 7.6%) compared to ChatGPT. This reflects Claude’s use for harder, deeper tasks — more opportunities for both clear success and genuine difficulty.

Correction Density

Average corrections per conversation: 0.49 overall (ChatGPT: 0.39, Claude: 0.84).

Correction density increases over time — from near zero in 2023 to 1.14 in April 2026. This does *not* indicate declining prompt quality. It reflects a shift toward harder tasks: the early period was dominated by short Q&A conversations; the recent period involves complex systems engineering (a major AI project, The Muser) where iterative refinement is the expected workflow, not a failure mode. The 4.3x higher rate on Claude versus ChatGPT supports this interpretation — Claude is used for the most technically demanding work.

Efficiency

The median output-to-input word ratio is 12.3x — for every word the user writes, the assistant produces 12.3 words. ChatGPT is wordier (14.0x median) compared to Claude (7.3x), suggesting either that Claude produces more concise responses or that Claude conversations involve more user context (higher word input per message), or both.

Session Dynamics

Session Type	Count	Percentage
Sprint (<30 min)	1,318	56.0%
Working (30 min – 2 hr)	398	16.9%
Marathon (>2 hr)	455	19.3%
Multi-day (>24 hr)	180	7.6%

19.3% of conversations are marathon sessions exceeding two hours — nearly one in five. Combined with the 7.6% multi-day sessions, over a quarter of all conversations represent sustained, project-level engagement rather than quick lookups.

4. Interaction Patterns

4.1 Conversation Control

Style	Percentage
Informational (providing context/data)	57.2%
Interrogative (asking questions)	27.3%
Feedback (evaluating responses)	11.1%
Directive (imperative instructions)	3.3%
Collaborative (“let’s...”)	1.2%

The dominant interaction style is **informational** — the user provides context, data, or specifications rather than issuing commands. This is characteristic of someone who has learned that giving the model more context produces better results than giving it more instructions. The low directive rate (3.3%) and high informational rate (57.2%) suggest a working style that treats the LLM as a collaborator requiring briefing, not an executor requiring orders.

4.2 Usage Modes

Mode	Conversations	Percentage
Mixed (no dominant mode)	2,074	87.8%
Teaching/Learning	98	4.2%
Synthesis (comparing/combining)	75	3.2%
Production (generating deliverables)	61	2.6%
Exploration (broad, short queries)	53	2.2%

The high “mixed” rate (87.8%) indicates that most conversations blend multiple modes — asking questions, providing context, requesting output, and evaluating results within a single session. This is consistent with iterative, project-driven work rather than single-purpose queries.

5. Professional Readiness Assessment

5.1 Demonstrated Competencies

Each competency below is backed by specific metrics from this analysis:

Multi-model proficiency. 21 distinct model versions used across the full GPT lineage, plus reasoning models and deep research. Each new model generation adopted within days of release. This is not theoretical knowledge of model capabilities — it is demonstrated, sustained use across the full frontier model landscape.

Platform versatility. Concurrent use of ChatGPT and Claude with demonstrably different usage patterns per platform — deeper conversations, more code inclusion, and more structured prompting on Claude. 22 Claude projects with prompt templates and attached documents show systematic project organization.

Feature adoption. Active use of 17 tool and feature types including web search, code interpreter, extended reasoning, vision, image generation, canvas, and deep research. 331 custom GPT conversations across 35 configurations demonstrate ecosystem engagement.

Prompt technique maturity. Measurable adoption of constraint specification (6.5%), context front-loading (4.6%), iterative refinement (3.3%), code-integrated prompting (2.7%), multi-step instructions (1.7%), and example provision (1.3%) — with higher rates on the more technically demanding platform.

Sustained engagement. 883 active days across 33 months (88% of all days). This is not a burst of enthusiasm — it is a daily practice sustained for nearly three years.

Project delivery. Conversations directly linked to completed outputs: a major AI pipeline project (100,000+ lines of Python, hundreds of processed works), VCAT Voynich data project (4,072 records, 343 tests, published on GitHub and HuggingFace), published technical articles, completed academic coursework.

5.2 Comparative Context

The scale of this usage places it well outside typical individual consumption:

- 71,427 messages across 2,398 conversations over 33 months
- 23.4 million tokens exchanged (equivalent to ~234 full-length novels)
- 2,061,417 words written in prompts alone (~20 novels of authored input)
- Top 1% ChatGPT user by messages sent (OpenAI, 2025)
- Documented in-chat practice time: 3,737 hours (Tier 1 auditable baseline); total LLM-assisted practice: 6,465 hours (Tier 2 estimated baseline)

For reference, the widely cited 10,000-hour threshold for expert-level practice represents approximately 5 years of full-time work. The Tier 2 estimate of 6,465 hours, accumulated over 33 months, represents approximately 64.6% of that threshold — achieved in a domain that has only existed in its current form since late 2022.

Methodology and Limitations

Data Sources

All data derives from complete official exports obtained directly from OpenAI (ChatGPT) and Anthropic (Claude) via their platform data export features. No sampling, estimation, or extrapolation was applied to the raw data.

- **ChatGPT export** (April 22, 2026): 19 JSON shards totaling ~298 MB, covering August 2023 – April 2026
- **Claude export** (April 21, 2026): conversations.json (~284 MB) + projects.json (~6.5 MB), covering March 2024 – April 2026

Classification Methodology

Topic classification uses a two-tier keyword matching approach: Tier 1 matches against conversation titles; Tier 2 falls back to first-message content for ambiguous or missing titles. A 10% stratified validation sample showed 93.1% agreement between title-only and title+content classification. Four categories were flagged with lower confidence: AI/ML/LLM

(68.8% agreement), Systems/Hardware/DevOps (70.0%). Two categories (Military/Strategy, Voynich/Cryptography) had too few samples for reliable validation.

The 46.4% “Other” rate reflects the inherent limitation of keyword matching on short or colloquial conversation titles. An embedding-based or LLM-assisted classifier would likely reduce this substantially but was excluded to maintain full offline reproducibility.

Project attribution uses keyword matching against known project names and descriptions. Confidence levels are explicitly reported: “high” requires the project name in the title; “medium” requires 2+ keyword matches; “low” requires a single keyword match. Claude project-to-conversation associations could not be extracted from the export (the `chat_conversations` field is empty for all 22 projects), so all Claude attribution is keyword-based.

Technique Detection

Prompt engineering techniques are detected via compiled regular expressions applied to user message text. This approach reliably detects explicit structural markers (code fences, numbered lists, “you are a...” phrases) but cannot assess prompt quality, specificity, or appropriateness. A message containing “do not” is counted as constraint-setting whether the constraint is well-designed or trivial. The technique adoption rates are most meaningful in their relative trends over time and cross-platform comparisons, not as absolute quality indicators.

Conversation Outcome Heuristics

The convergence/frustration classification uses end-of-conversation signals and correction counts. “Frustrated” requires 3+ correction-pattern messages or explicit frustration language. “Converged” requires explicit positive feedback in the last user message. The dominant “standard completion” category (82%) represents conversations that ended normally without explicit positive or negative signals — the absence of frustration should not be read as the presence of satisfaction, nor vice versa. These heuristics provide directional signals, not definitive outcome measurement.

Correction Density Interpretation

Rising correction density over time correlates with increasing task complexity (shift from Q&A to systems engineering), not declining prompt quality. The 4.3x higher correction rate on Claude versus ChatGPT supports this: Claude is used for the most demanding work where iterative refinement is the expected workflow.

Reproducibility

All analysis scripts are located at /analysis/deep-analysis/scripts/ and produce their outputs from the raw export data. The scripts, their inputs, and their outputs are sufficient to reproduce every figure in this document.

Appendices

A. Tool and Feature Usage Detail

Feature	Conversations	% of ChatGPT
Web Search	473	25.8%
Code Interpreter	383	20.9%
Extended Reasoning	283	15.4%
Multimodal/Vision	246	13.4%
Memory	137	7.5%
Image Generation	72	3.9%
File Browser	39	2.1%
File Search	32	1.7%
Canvas	13	0.7%
Deep Research	2	0.1%
Generated UI	1	0.1%

B. Branching and Regeneration

213 conversations (11.6% of ChatGPT total) contain at least one branch point — indicating the user regenerated a response or explored an alternative conversation path. Total branch points across all conversations: 341. Distribution: 150 conversations have exactly 1 branch, with a steep dropoff thereafter.

C. Finish Details

Outcome	Percentage
Normal completion (stop)	97.43%

Outcome	Percentage
User interrupted	2.51%
Max tokens reached	0.04%
Content filter	0.03%

D. Custom GPT Ecosystem

331 conversations used custom GPTs across 35 unique template configurations. This represents 18.0% of all ChatGPT conversations and indicates active engagement with the custom GPT ecosystem for specialized workflows.

E. Session Type Distribution

Session Type	Count	Percentage
Sprint (<30 min)	1,318	56.0%
Working (30 min – 2 hr)	398	16.9%
Marathon (>2 hr)	455	19.3%
Multi-day (>24 hr)	180	7.6%